Exercise 9.1. *Solve the Cauchy-Euler equation (9.8) to obtain the solution (9.9).*

$$\frac{d^2R}{d\rho^2}+\frac{1}{\rho}\frac{dR}{d\rho}-\frac{m^2}{\rho^2}R=0 \qquad (9.8)$$

级数解 $R(\rho)=\sum_n a_n\rho^n$ $\quad \frac{dR}{d\rho}=\sum_n n a_n \rho^{n-1}$ $\quad \frac{d^2R}{d\rho^2}=\sum_n n(n-1)a_n\rho^{n-2}$ 代回

逐项系数相等 $n(n-1)+n-m^2=0 \qquad n^2=m^2$

$m=0$ 时, $R(\rho)=E+F\ln\rho$

$m\in\mathbb{N}$ 时, $n=\pm m$, $R(\rho)=E\rho^m+F\rho^{-m}$

Exercise 9.7. *Verify that the power-series representation of the Bessel functions satisfy the Recursive formulae (9.61) and (9.62).*

$$2J_n'(\zeta)=J_{n-1}(\zeta)-J_{n+1}(\zeta), \qquad (9.61)$$

$$\frac{2n}{\zeta}J_n(\zeta)=J_{n-1}(\zeta)+J_{n+1}(\zeta). \qquad (9.62)$$

$$J_{n-1}(\zeta)-2J_n'(\zeta)=\sum_{l=0}^{\infty}\frac{(-1)^l}{l!(l+n)!}(l+n)\left(\frac{\zeta}{2}\right)^{2l+n-1}-\sum_{l=0}^{\infty}\frac{(-1)^l}{l!(l+n)!}(2l+n)\left(\frac{\zeta}{2}\right)^{2l+n-1}$$

$$=\sum_{l=0}^{\infty}\left(\frac{\zeta}{2}\right)^{2l+n-1}\frac{(-1)^l(-l)}{l!(l+n)!}$$

$$=\frac{2n}{\zeta}J_n(\zeta)-J_{n-1}(\zeta)$$

$$=\sum_{l=0}^{\infty}\left[\frac{(-1)^l n}{l!(l+n)!}\left(\frac{\zeta}{2}\right)^{2l+n-1}-\frac{(-1)^l}{l!(l+n-1)!}\left(\frac{\zeta}{2}\right)^{2l+n-1}\right]$$

$$=\sum_{l=0}^{\infty}\left(\frac{\zeta}{2}\right)^{2l+n-1}\frac{(-1)^l(-l)}{l!(l+n)!}\overset{l=0\text{时}}{=0}=\sum_{l=1}^{\infty}\frac{(-1)^{l+1}}{(l-1)!(l+n)!}\left(\frac{\zeta}{2}\right)^{2l+n-1}$$

$$=\sum_{l=0}^{\infty}\frac{(-1)^{l+2}}{l!(l+n+1)!}\left(\frac{\zeta}{2}\right)^{2l+n+1}=J_{n+1}(\zeta)$$

Exercise 9.10. *Rederive the recursive formulae (9.61) and (9.62) of Bessel functions using the method introduced in this subsection.*

$$\langle\rho,\varphi|P_\pm|\rho,m\rangle=\langle\rho,\varphi|P_\pm|\rho,m\rangle$$

$$\int_0^\infty\int_0^{2\pi}\langle\rho,\varphi|P_\pm|\rho',\varphi'\rangle\langle\rho',\varphi'|\rho,m\rangle\rho'd\rho'\frac{d\varphi'}{2\pi}=\sum_{m'\in\mathbb{Z}}\langle\rho,\varphi|\rho,m'\rangle\langle\rho,m'|P_\pm|\rho,m\rangle$$

$$-ie^{\pm i\varphi}\left(\frac{\partial}{\partial\rho}\pm\frac{i}{\rho}\frac{\partial}{\partial\varphi}\right)\langle\rho,\varphi|\rho,m\rangle=\mp\rho\langle\rho,\varphi|\rho,m\pm1\rangle$$

$$\langle\rho,\varphi|\rho,m\rangle=J_m(\rho)e^{im\varphi}$$

$$\Rightarrow J_n'(\zeta)\mp\frac{n}{\zeta}J_n(\zeta)=\mp J_{n\pm1}(\zeta)$$

Exercise 9.13. *Show that*

$$[N, a_\pm] = \pm a_\pm.$$

$$[N, a_+] = [a_+a_-, a_+] = a_+[a_-, a_+] = a_+ \qquad [N, a_-] = [a_+, a_-]a_- = -a_-$$

Exercise 9.16. *Use the equation above and what you have obtained in Exercise 9.15 to find the addition formula of the Hermite polynomials.*

$$e^{2t\xi - t^2} = \sum_{n=0}^{\infty} H_n(\xi)\frac{t^n}{n!} \qquad e^{2t\xi_1 - t^2} e^{2t\xi_2 - t^2} = \sum_{m,n=0}^{\infty} H_m(\xi_1)H_n(\xi_2)\frac{t^{m+n}}{m!\,n!}$$

$$e^{2t(\xi_1+\xi_2) - 2t^2} = e^{2\cdot\sqrt{2}t\,\frac{1}{\sqrt{2}}(\xi_1+\xi_2) - (\sqrt{2}t)^2} = \sum_{n=0}^{\infty} H_n\left[\frac{1}{\sqrt{2}}(\xi_1+\xi_2)\right]\frac{(\sqrt{2}t)^n}{n!}$$

Exercise 9.22. *Prove the identities*

$$\varepsilon^{jkl}\varepsilon^{jmn} = \delta^{km}\delta^{ln} - \delta^{kn}\delta^{lm}, \tag{9.132}$$

$$\varepsilon^{jkl}\varepsilon^{jkm} = 2\delta^{lm}, \tag{9.133}$$

$$\varepsilon^{jkl}\varepsilon^{jkl} = 6. \tag{9.134}$$

$\varepsilon^{jkl}\varepsilon^{jmn}$ $\quad k=m \ l=n$ 时 $\varepsilon^{jkl} = \varepsilon^{jmn}$ $\quad k=n \ l=m$ 时 $\varepsilon^{jkl} = -\varepsilon^{jmn}$ $\quad k,l,m,n=j$ 时, $=0$

$$\varepsilon^{jkl}\varepsilon^{jkm} = \delta^{kk}\delta^{lm} - \delta^{km}\delta^{lk} = 3\delta^{lm} - \delta^{lm} = 2\delta^{lm}$$

$$\varepsilon^{jkl}\varepsilon^{jkl} = \delta^{kk}\delta^{ll} - \delta^{kl}\delta^{lk} = 9-3=6$$

Exercise 9.25. *Verify that* $\vec{J}^2|m\rangle = j(j+1)|m\rangle$, $-j < m < j$.

$[\vec{J}^2, J_\pm] = 0 \qquad \vec{J}^2|j\rangle = j(j+1)|j\rangle \qquad |m\rangle \propto J_\pm^k|j\rangle \qquad \vec{J}^2|m\rangle = j(j+1)|m\rangle$

Exercise 9.28. *Similarly, show that a symmetric tensor* $T^{\{kl\}}$ *also remains symmetric under* $SO(3)$ *transformation.*

$$T^{\{kl\}} \to R^{mk}R^{nl}T^{\{kl\}} = \frac{1}{2}R^{mk}R^{nl}(T^{kl}+T^{lk}) = \frac{1}{2}(R^{mk}R^{nl}T^{kl} + R^{mk}R^{nl}T^{lk})$$

$$= \frac{1}{2}(T'^{mn} + T'^{nm}) = T'^{\{mn\}}$$

Exercise 9.31. *Verify that* $\tilde{S}^{jkl}$ *is indeed traceless. Then you would understand where the factor* $\frac{1}{5}$ *comes from.*

$$\mathrm{Tr}\,\tilde{S}^{jkl} = \tilde{S}^{jkl}\delta_{jk} = \left[S^{jkl} - \frac{1}{5}(\delta^{jk}u^l + \delta^{kl}u^j + \delta^{lj}u^k)\right]\delta_{jk}$$

$$= S^{jkl}\delta_{jk} - \frac{1}{5}(3u^l + u^l + u^l)$$

$$= u^l - u^l = 0$$

Exercise 9.34. *Explicitly rewrite $\tilde{T}^{\{jk\}}V^l$ in terms of $\tilde{S}$, U, and X.*

$$X^{jm} = \frac{1}{2}(W^{jm} + W^{mj}) \qquad W^{jm} = \tilde{T}^{\{jk\}} V^l \varepsilon^{klm}$$

$$\tilde{S}^{jkl} = S^{jkl} - \frac{1}{5}(\delta^{jk} U^l + \delta^{kl} U^j + \delta^{lj} U^k)$$

$$U^l = \delta^{jk} S^{jkl} = 2\tilde{T}^{\{jl\}} V^j$$

Exercise 9.37. *Use the orthonormality of the basis vectors to show that*

$$\sum_{J,M} C^{JM}_{j_1 j_2 m_1 m_2} C^{j_1 j_2 m_1' m_2'}_{JM} = \delta^{m_1'}_{m_1} \delta^{m_2'}_{m_2}, \tag{9.152}$$

$$\sum_{m_1,m_2} C^{JM}_{j_1 j_2 m_1 m_2} C^{j_1 j_2 m_1 m_2}_{J'M'} = \delta^{J}_{J'} \delta^{M}_{M'}. \tag{9.153}$$

$$\sum_{J,M} C^{JM}_{j_1 j_2 m_1 m_2} C^{j_1 j_2 m_1' m_2'}_{JM} = \sum_{JM} \langle j_1, m_1; j_2, m_2 | J, M \rangle \langle J, M | j_1, m_1'; j_2, m_2' \rangle$$

$$= \langle j_1, m_1; j_2, m_2 | j_1, m_1'; j_2, m_2' \rangle = \delta^{m_1'}_{m_1} \delta^{m_2'}_{m_2}$$

$$\sum_{m_1,m_2} C^{JM}_{j_1 j_2 m_1 m_2} C^{j_1 j_2 m_1 m_2}_{J'M'} = \sum_{m_1,m_2} \langle j_1, m_1; j_2, m_2 | J, M \rangle \langle J', M' | j_1, m_1; j_2, m_2 \rangle$$

$$= \sum_{m_1,m_2} \langle J', M' | j_1, m_1; j_2, m_2 \rangle \langle j_1, m_1; j_2, m_2 | J, M \rangle$$

$$= \langle J', M' | J, M \rangle = \delta^{J}_{J'} \delta^{M}_{M'}$$

Exercise 9.40. *Verify that $\{X_-, X_3, X_+\}$ are indeed vector operators.*

$J_3 = (\vec{x} \times \vec{p})_3 = x_1 p_2 - x_2 p_1 \qquad [x_i, p_j] = i\delta_{ij} \qquad \hbar \equiv 1$

$J_2 = x_3 p_1 - x_1 p_3 \qquad J_1 = x_2 p_3 - x_3 p_2 \qquad J_+ = J_1 + iJ_2 = x_2 p_3 - x_3 p_2 + i(x_3 p_1 - x_1 p_3)$

$X_- = X_1 - iX_2 \qquad X_+ = X_1 + iX_2 \qquad J_- = J_1 - iJ_2 = x_2 p_3 - x_3 p_2 - i(x_3 p_1 - x_1 p_3)$

$[J_3, x_3] = 0$

$[J_+, x_3] = x_2 [p_3, x_3] - i x_1 [p_3, x_3] = -x_1 - i x_2 = -X_+$

$[J_-, x_3] = x_2 [p_3, x_3] + i x_1 [p_3, x_3] = x_1 - i x_2 = X_-$

$[J_3, X_+] = -x_2 [p_1, x_1] + i x_1 [p_2, x_2] = X_+$

$[J_3, X_-] = -x_2 [p_1, x_1] - i x_1 [p_2, x_2] = -X_-$

Exercise 9.43. *Show that the electric quadruple moments are indeed $j = 2$ tensor operators.*

$$Q_{k\ell} = \int_{\mathbb{R}^3} \rho(\vec{x}) \left(3 x_k x_\ell - |\vec{x}|^2 \delta_{k\ell}\right) d^3\vec{x}$$

$$[x_1^2 + x_2^2 + x_3^2, p_1] = x_1[x_1, p_1] + [x_1, p_1] x_1 = 2i x_1$$

$$[J_3, Q_{k\ell}] = \int_{\mathbb{R}^3} \rho(\vec{x}) \; 3\left(x_1[p_2, x_k x_\ell] - x_2[p_1, x_k x_\ell]\right)$$

$x_1 p_2 - x_2 p_1$

$$- \delta_{k\ell} \left(x_1[p_2, |\vec{x}|^2] - x_2[p_1, |\vec{x}|^2] \right) d^3\vec{x}$$

$$= \int_{\mathbb{R}^3} \rho(\vec{x}) \; -3i \left[\delta_{k2} x_1 x_\ell + \delta_{k2} \delta_{\ell 2} x_1 x_2 \right.$$

Exercise 9.46. *What are the homotopy groups of $SO(3)$?*

$\pi_0(SO(3)) = 0 \quad \pi_1(SO(3)) = \mathbb{Z}_2 \quad \pi_2(SO(3)) = 0 \quad \pi_3(SO(3)) = \mathbb{Z} \quad \pi_k(SO(3)) = 0,\ k>3$

Exercise 9.52. *Show that*

$$d^j(\beta)_{m'm} = (-1)^{m-m'} d^j(\beta)_{mm'}$$

for all j, m, m'. Hint: First show that J_2 is an antisymmetric matrix in the basis $|j,m\rangle$; then check the symmetry properties of the matrices J_2^k in this basis for k even and odd; and finally use the exponential map.

$J_2 = \frac{1}{2i}(J_+ - J_-) \qquad \hbar = 1$

$\langle j,m|J_2|j,m'\rangle = \frac{1}{2i}\left(\delta_{m',m+1}\sqrt{(j-m)(j+m+1)} - \delta_{m',m-1}\sqrt{(j+m)(j-m+1)}\right) = -(J_2)_{m'm}$

k even $\quad J_2^T = -J_2 \quad (J_2^k)_{m'm} = (J_2^k)_{mm'}$

k odd $\quad (J_2^k)_{m'm} = -(J_2^k)_{mm'}$

$$d^j(\beta)_{m'm} = \langle j,m'|e^{-i\beta J_2}|j,m\rangle = \sum_{k=0}^{\infty} \frac{(-i\beta)^k}{k!}(J_2^k)_{m'm}$$

$$= \sum_{k\in\text{even}} \frac{(-i\beta)^k}{k!}(J_2^k)_{mm'} - \sum_{k\in\text{odd}} \frac{(-i\beta)^k}{k!}(J_2^k)_{mm'}$$

$$= \sum_{k\in\text{even}} \frac{(i\beta)^k \overbrace{(-1)^k}^{=1}}{k!}(J_2^k)_{mm'} + \sum_{k\in\text{odd}} \frac{(i\beta)^k \overbrace{(-1)^{k+1}}^{=1}}{k!}(J_2^k)_{mm'}$$

$$= \sum_k \frac{(i\beta)^k}{k!}(J_2^k)_{mm'} = \langle j,m|e^{i\beta J_2}|j,m'\rangle$$

$$= (-1)^{m-m'} d^j(\beta)_{mm'}$$

$$(-1)^{m-m'} \sum_{k=0}^{\infty} \frac{(-1)^k}{k!}(J_2^k)_{mm'}$$

k odd $\quad (-1)^{m-m'+1}$

k even $\quad (-1)^{m-m'}$

Exercise 9.61. *In the Euler-angle parameterization, show that the orthogonality (9.215) reduces to*

$$\frac{2j+1}{2}\int_0^\pi \mathrm{d}\beta \sin\beta d^j(\beta)_{mn} d^{j'}(\beta)_{mn} = \delta^{jj'}, \tag{9.219}$$

where m, n are not summed.

$$(2j+1)\int \mathrm{d}\tilde{\mu}(R) D^j(R)^*_{m'm} D^{j'}(R)_{n'n} = \delta^{jj'}\delta_{m'n'}\delta_{mn},$$

$$d\tilde{\mu}(R) = \frac{1}{8\pi^2}\sin\beta\, d\alpha\, d\beta\, d\gamma \qquad D^j(R)_{m'm} = e^{-i\alpha m' - i\gamma m} d^j(\beta)_{m'm}$$

$$\frac{2j+1}{2}\int_0^\pi \sin\beta\, d\beta \int_0^{2\pi}\int_0^{2\pi} d\alpha\, d\gamma\, \frac{1}{4\pi^2} d^j(\beta)_{m'm} d^{j'}(\beta)_{n'n} = \delta^{jj'}\delta_{m'n'}\delta_{mn}$$

$$\Rightarrow \frac{2j+1}{2}\int_0^\pi d\beta \sin\beta\, d^j(\beta)_{mn} d^{j'}(\beta)_{mn} = \delta^{jj'}$$

Exercise 9.64. *Prove the symmetry property of spherical harmonics*

$$Y^l_{-m}(\theta,\phi) = (-1)^m Y^l_m(\theta,\phi)^*.$$

$$Y^l_m(\theta,\phi) = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}\, P^l_m(\cos\theta)\, e^{im\phi} \qquad P^l_{-m}(x) = (-1)^m \frac{(l-m)!}{(l+m)!} P^l_m(x)$$

$$(-1)^m Y^l_m(\theta,\phi)^* = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}\, P^l_m(\cos\theta)\, e^{-im\phi}(-1)^m$$

$$Y^l_{-m}(\theta,\phi) = \sqrt{\frac{2l+1}{4\pi}\frac{(l+m)!}{(l-m)!}}\, P^l_{-m}(\cos\theta)\, e^{-im\phi}$$

对比可得.

Exercise 9.67. *Find the associated Legendre equation using the differential equation of the Wigner d-functions.*

$$\left[\frac{d^2}{d\beta^2} + \cot(\beta)\frac{d}{d\beta} + \left(j(j+1) - \frac{m^2}{\sin^2\beta} - \frac{m'^2}{\cos^2\beta}\right)\right] d^j_{mm'}(\beta) = 0$$

$$x = \cos\beta \qquad \frac{d}{d\beta} = -\sin\beta\frac{d}{dx} \qquad \frac{d^2}{d\beta^2} = \sin^2\beta\frac{d^2}{dx^2} + \cos\beta\frac{d}{dx} \qquad \cot\beta\frac{d}{d\beta} = -x\frac{d}{dx}$$

$$\Rightarrow \left[(1-x^2)\frac{d^2}{dx^2} - 2x\frac{d}{dx} + j(j+1) - \frac{m^2}{1-x^2}\right] P^m_l(x) = 0$$